集合论

Question 1

$A = \{f : N \to N : \exists N \in N, \forall n \geq N, f(n) = 0\}$ and $B = \{f : N \to N : \forall N \in N, \exists n \geq N, f(n) = 0\}$.

Show that $A \subseteq B$

Show that the inclusion in part (a) is strict; namely, that $A \neq B$.

If $f, g : N \to N$ are functions, we define their sum $f + g : N \to N$ as $(f + g)(n) = f(n) + g(n)$ for all $n \in N$. Show that A is closed under addition; that is, if $f, g \in A$ then $f + g \in A$. Be sure to justify your answer

With addition defined as in part (iii), is B closed under addition? Be sure to justify your answer.

数学归纳法

Question 2

Consider the following set $S \subseteq Z$, which we define recursively:

- $0 \in S$
- If $n \in S$ then $2n + 1 \in S$.

Write out six (6) elements of this set.

Prove that $s \in S$ if and only if $s = 2^n - 1$ for some non-negative integer n.

Question 3

Recall that the Fibonacci sequence is defined as $F_1 = F_2 = 1$ with $F_{n+1} = F_n + F_{n-1}$ for all $n \geq 2$. Show that for all $n \geq 10$.

$$\left(\frac{3}{2}\right)^n < F_n < 2^n$$

Hint: You may free to use the fact that $\left(\frac{3}{2}\right)^{10} < 87$

关系

To distinguish between different types of relations, we define properties that the relation can exhibit. If R is a relation on S then

- R is **reflexive** if for all a, aRa,
- R is **irreflexive** if for all $a \in A$, $\sim aRa$.
- R is **transitive** if whenever aRb and bRc then aRc
- R is **symmetric** if whenever aRb then bRa
- R is **anti-symmetric** if whenever aRb and bRa, then $a = b$
- R is **total** if for all a and b either aRb or bRa.
- R is **left-total** if for every $a \in A$ there exists $b \in B$ such that aRb.

Weak order Relation

A relation R on a set A is said to be a **weak order relation** if R is transitive, anti-symmetric, and reflexive.

Strong order Relation

A relation R on a set A is said to be a **strong order relation** if R is transitive, anti-symmetric, and irreflexive.

Total Order

We say that R is a **linear or total order** on A if it is total

Partial Order

We say that R is a **Partial Order** on A if it is not total

Equivalence Relation（等价关系）

An relation R **on a set** S is said to be an equivalence relation if R is **reflexive**, **transitive**, and **symmetric**.

Question 4

Suppose that $A \subseteq N$ is a non-empty set, and define a relation $\sim$ on A as

$$a \sim b \text{ if and only if } a \mid b \text{ or } b \mid a.$$

Show that for any set A, $\sim$ is symmetric and reflexive.

Determine whether $\sim$ is an equivalence relation on $A = N$.

Determine whether $\sim$ is an equivalence relation on $A = \{2^n : n \in N\}$.

Question 5

Let $E \subseteq (0, \infty)$ be a nonempty set satisfying the following properties:

(E1) $1 \notin E$.

(E2) If $u, v \in E$, then both $uv \in E$ and $u^v \in E$.

(E3) For every real number $t > 0$ with $t \neq 1$, exactly one of $t \in E$ or $\frac{1}{t} \in E$ holds (but not both).

You may assume that such a set E exists. Define a relation ▷ on $(0, \infty)$ by x▷y if and only if $y/x \in E$.

Prove that ▷ is a strict partial order (irreflexive, transitive, and anti-symmetric).

Prove that if $x \triangleright y$ and $r > 1$, then $x^r \triangleright y^r$. Hint: Break this into the cases where $r \in E$ and $\frac{1}{r} \in E$.

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Question 6

Suppose that S is a set. Given a function $f : S \to S$ and a positive integer n, we define

$$f^{\circ n} = f \circ f \circ \ldots . f \ (n\ times)$$

Suppose that $f : S \to S$ satisfies $f^{\circ n} = f$ for some integer $n \geq 2$. Show that f is injective if and only if it is bijective. Is the same statement true if we replace "injective" with "surjective"?

Question 7

Suppose that $f : X \to Y$ is an injective function. If $A, B \subseteq X$ and $f(A) = f(B)$, show that $A = B$.

基数

Definition: countable set.

A set S is said to be countable if $|S| \leq |N|$

Definition: countable Infinite.

A set S is said to be countable infinite if $|S| = |N|$

Definition: uncountable set.

An infinite set that is not countable, is called an **uncountable** set.

Theorem:

If S and T are sets, then $|S| = |T|$, if there is a bijection $S \to T$.

Theorem: **Schr¨oder-Bernstein Theorem**

If there is two injection $f: A \to B$, 有一个$g: B \to A$, then there is a bijection

$$h: A \to B$$

当然 surjective 也成立，但是老师没写

Theorem:

A countable union of (pairwise disjoint) countable sets is countable.

Theorem: Cantor’s Theorem

If S is a set, then $|S| < |P(S)|$.

If S is a finite set, then

$$|P(S)| = 2^{|S|}$$

Theorem: Dimension and Surjectivity

Suppose that V and W be two sets

If $|V| < |W|$ then $T : V \to W$ is not surjective.

Note:

The contrapositive is also helpful: If $T : V \to W$ is surjective the $|W| \leq |V|$.

Theorem: Dimension and Injectivity

Suppose that V and W be two sets

If $|W| < |V|$then $T : V \to W$ is not Injective.

Note:

The contrapositive is also helpful: If $T : V \to W$ is injective the $|V| \leq |W|$.

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Question 8

Suppose that A, B, and C are sets such that $|A| = |B|$ and $|B| \leq |C|$. Prove that $|A| \leq |C|$.

Question 9

Recall that for any sets S and T, we define $|S| + |T| = |S \times \{0\} \cup T \times \{1\}|$. Show that

$$|S| + |T| \geq |S \cup T|.$$

Suppose that S and T are finite sets such that $|S| + |T| > |S \cup T|$. Show that $S \cap T \neq \emptyset$.

Suppose that $n \geq 2$ is a positive integer and A is a subset of $\{1, 2, \ldots, n-1\}$ such that $|A| \geq \frac{n}{2}$. Show that there exist $a_1, a_2 \in A$ such that $a_1 + a_2 = n$. Hint: Use part (ii), where one of your sets is $T = \{n - a : a \in A\}$.

Question 10

Let G and H be groups and $\phi: G \to H$ be a group homomorphism.

Prove that $ker(\phi) \leq G$.

A subgroup $N \leq G$ is said to be normal if $gng^{-1} \in N$ for all $g \in G$ and $n \in N$. Show that $ker(\phi)$ is a normal subgroup of G.

Question 11

Suppose that G is a group in which $(ab)^2 = a^2b^2$ for all $a, b \in G$. Show that G is abelian.

数论

Definition: Divide

If $a, b \in Z$ with $b \neq 0$, and $a = mb$ for some integer m, then a is divisible by b, or we sat b divides a (written as $b \mid a$).

We call b a divisor or factor of a

Theorem: Euclid' s Lamma

Let $p > 1$ be a prime number, and $a, b \in Z$. If $p|ab$, then $p|a \; or \; p|b$.

Theorem:

If $a|bc$, and $\gcd(a, b) = 1$, then $a|c$

Theorem: (不太常用)

If $a, b, c \in Z$, then $ax + by = c$ has a solution if and only if $gcd(a, b) \, |c$.

Theorem: The Division Algorithm

If $a, b \in N$, then there is a unique pair of integers, q and r , with $q \geq 0$ and $0 \leq r < b$ such that $a = q \times b + r$.

Definition: GCD

If $a, b \in Z$, we define the greatest common divisor of a and b, written $gcd(a, b)$, to be the largest positive integer that divides both a and b. More precisely,

$$gcd(a, b) = max\{d \in Z : d > 0 \; and \; d|a \; and \; d|b\}.$$

Theorem: (这个要用得稍微证一下)

If $\gcd(a, b) = d$, then $\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$

Theorem: Bezout' s Identity

Let a and b be to Integers. Then there are $m, n \in Z$, such that

$$a * m + b * n = \gcd(a, b)$$

Theorem: (不太常用，是欧几里得算最大公因数的理论基础)

If $a, b, q, r \in Z$ are such that $a = bq + r$, then $gcd(a, b) = gcd(b, r)$.

Theorem:

Suppose that $a, b, c \in Z$ and (x_0, y_0) satisfy $ax_0 + by_0 = c$. If $d = gcd(a, b) \neq 0$ then the general solution to the Diophantine equation $ax + by = c$ is given By

$$x = x_0 + \frac{nb}{d}, y = y_0 - \frac{na}{d}$$

Definition: Prime Number

A natural number greater than 1 is prime if its only positive factors is itself and 1.

Definition: Relative Prime

If $GCD(a, b) = 1$, we say that a and b are relative prime.

Theorem:

There are infinitely many prime numbers.

Definition: Rational Numbers: $\mathbb{Q}$

有理数是可以表示为两个整数的商的数，其中分母不为零

$$Q = \{\frac{p}{q} | p, q \in Z, q \neq 0\}$$

Theorem: The Fundamental Theorem of Arithmetic

Every Natural Number $n \geq 2$, is either a prime, or can be expressed as a product of distinct primes, in a unique way.

Definition: Mod

If $n \in Z^+$ and $a, b \in Z$, we say that $a \equiv b \ (mod\ n)$ (read: a is congruent to b mod n) if $n \mid (b - a)$.

Theorem:

Fix an $n \in Z^+$. If $a \equiv r \ (mod\ n)$ and $b \equiv s \ (mod\ n)$ then

1. $(a + b) \equiv (r + s) \ (mod\ n)$.
2. $ab \equiv rs \ (mod\ n)$

Theorem : Fermat's Little Theorem

If $a \in Z, p \in N$ are such that p is prime and $p \nmid a$,then

$$a^{p-1} \equiv 1 \ (mod\ p)$$

Theorem

For any integer a and prime p, $a^p \equiv a \ (mod\ p)$.

Theorem

If p is prime, then every non-zero element of $\mathbb{Z}_p$ has a multiplicative inverse.

Question 12

For a positive natural number n, define $n!$ to be $n! = 1 \cdot 2 \cdot 3 \cdot \cdots \cdot n$, the product of the first n positive natural numbers.

Let p and q be distinct prime numbers, and set $d = pq$. Find the smallest n such that $d|n!$.

Determine the remainder when $15!$ is divided by 17.

Question 13

Suppose that $a, b, n \in Z^+$ and that $a \equiv b \ (mod\ n)$. Show that $gcd(a, n) = gcd(b, n)$.